

ident, *x* (* identifiers *)

<i>exp</i> , <i>e</i>	::=		expressions like <i>exp</i>
		<i>x</i>	variable named <i>x</i>
		()	unit value
		(<i>exp</i> , <i>exp'</i>)	pair of <i>exp</i> and <i>exp'</i>
		function <i>x</i> : type $\rightarrow$ <i>exp</i>	bind <i>x</i> in <i>exp</i>
		<i>exp exp'</i>	
		{ <i>exp</i> / <i>x</i> } <i>exp'</i>	M
		(<i>exp</i>)	M

type, <i>t</i>	::=	
		unit
		type $\star$ type'
		type $\rightarrow$ type'

<i>E</i>	::=	
		empty
		<i>E</i> , <i>x</i> : <i>t</i>
		<i>E</i> , <i>x</i> : <i>t</i> , <i>E'</i>

<i>val</i> , <i>v</i>	::=	
		<i>x</i>
		()
		(<i>val</i> , <i>val'</i>)
		function <i>x</i> : <i>t</i> $\rightarrow$ <i>exp</i>

<i>formula</i>	::=	
		<i>judgement</i>
		$\neg$ (<i>formula</i>)
		<i>x</i> $\in$ dom(<i>E</i>)
		M

<i>terminals</i>	::=	
		$\star$
		$\rightarrow$
		function
		-
		$\vdash$
		$\longrightarrow$
		{
		}

<i>Jtype</i>	::=		
		<i>E</i> $\vdash$ ok	M
		<i>E</i> $\vdash$ <i>x</i> : <i>t</i>	M
		<i>E</i> $\vdash$ <i>e</i> : <i>t</i>	
			term <i>e</i> has type <i>t</i> in type environment <i>E</i>

<i>Jop</i>	::=	
		<i>e</i> $\longrightarrow$ <i>e'</i>

<i>judgement</i>	::=	
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		$Jtype$
		Jop
$user_syntax$	$::=$	
		$ident$
		exp
		type
		E
		val
		$formula$
		$terminals$

$E \vdash \mathbf{ok}$

$\overline{\mathbf{empty} \vdash \mathbf{ok}}$	EOK_EMPTY
$\frac{E \vdash \mathbf{ok} \quad \neg(x \in \text{dom}(E))}{E, x : \mathbf{t} \vdash \mathbf{ok}}$	EOK_IDENT

$E \vdash x : \mathbf{t}$

$\frac{E, x : \mathbf{t}, E' \vdash \mathbf{ok}}{E, x : \mathbf{t}, E' \vdash x : \mathbf{t}}$	EXT_IDENT
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$E \vdash e : \mathbf{t}$

 term e has type $\mathbf{t}$ in type environment E

$\frac{E \vdash x : \mathbf{t}}{E \vdash x : \mathbf{t}}$	EET_IDENT
$\frac{E \vdash \mathbf{ok}}{E \vdash () : \mathbf{unit}}$	EET_UNIT
$\frac{E \vdash e_1 : \mathbf{t}_1 \quad E \vdash e_2 : \mathbf{t}_2}{E \vdash (e_1, e_2) : \mathbf{t}_1 \star \mathbf{t}_2}$	EET_PAIRPAIRING
$\frac{E, x : \mathbf{t}_1 \vdash e : \mathbf{t}_2}{E \vdash \mathbf{function} \ x : \mathbf{t}_1 \rightarrow e : \mathbf{t}_1 \rightarrow \mathbf{t}_2}$	EET_FUN
$\frac{E \vdash e : \mathbf{t}_1 \rightarrow \mathbf{t}_2 \quad E \vdash e' : \mathbf{t}_1}{E \vdash e \ e' : \mathbf{t}_2}$	EET_APP

$e \longrightarrow e'$

$\overline{(\mathbf{function} \ x : \mathbf{t} \rightarrow e) \ v \longrightarrow \{v/x\}e}$	JO_RED_FUN
$\frac{e_1 \longrightarrow e'_1}{e_1 \ e_2 \longrightarrow e'_1 \ e_2}$	JO_RED_CONTEXT1
$\frac{e_2 \longrightarrow e'_2}{v_1 \ e_2 \longrightarrow v_1 \ e'_2}$	JO_RED_CONTEXT2
$\frac{e_1 \longrightarrow e'_1}{(e_1, e_2) \longrightarrow (e'_1, e_2)}$	JO_RED_CONTEXT3

$$\frac{e_2 \longrightarrow e'_2}{(v_1, e_2) \longrightarrow (v_1, e'_2)} \quad \text{JO_RED_CONTEXT4}$$

Definition rules: 13 good 0 bad
Definition rule clauses: 27 good 0 bad